

WEATHER INFORMATION APP [GitHub](#)

Project Type: A Python-based application that provides **real-time weather information** for a selected city using a weather API.

Programming Language: Python

Internship: Python Developer Internship

Organization: Auspify Technologies

1. Description

The Weather Information App is a Python application that uses the Open Weather Map API to fetch current weather information.

The user enters a city name, and the application displays the city's temperature and current weather condition.

The application also handles invalid city names, connection problems, and empty input. Users can search for weather information for multiple cities without restarting the program.

2. Objective

The main objectives of this project are:

- To fetch real-time weather information using an API.
- To allow users to search for weather by city name.
- To display temperature and weather conditions.

- To learn how to work with REST APIs.
 - To process JSON data in Python.
 - To handle API and connection errors.
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3. Features

- City Search
 - Real-Time Temperature
 - Weather Condition
 - API Integration
 - Search Multiple Cities
 - API Error Handling
 - Empty Input Validation
 - Secure API Key Storage using .env
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4. Technologies Used

Technology	Purpose
Python	Develop the application
Open Weather Map API	Fetch weather data
Requests	Send API requests
JSON	Process weather data

Python-dotenv	Load API key securely
Console	User interaction

5. How It Works

The application works in the following steps:

1. The application loads the API key from the .env file.
 2. The user enters a city name.
 3. The application sends a request to the weather API.
 4. The API returns weather data in JSON format.
 5. The application extracts the temperature and weather condition.
 6. The information is displayed to the user.
 7. The user can search for another city or exit.
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6. How to Run

Follow these simple steps to run the Weather Information App:

Step 1: Install Python on your computer.

Step 2: Install the required libraries:

```
pip install requests python-dotenv
```

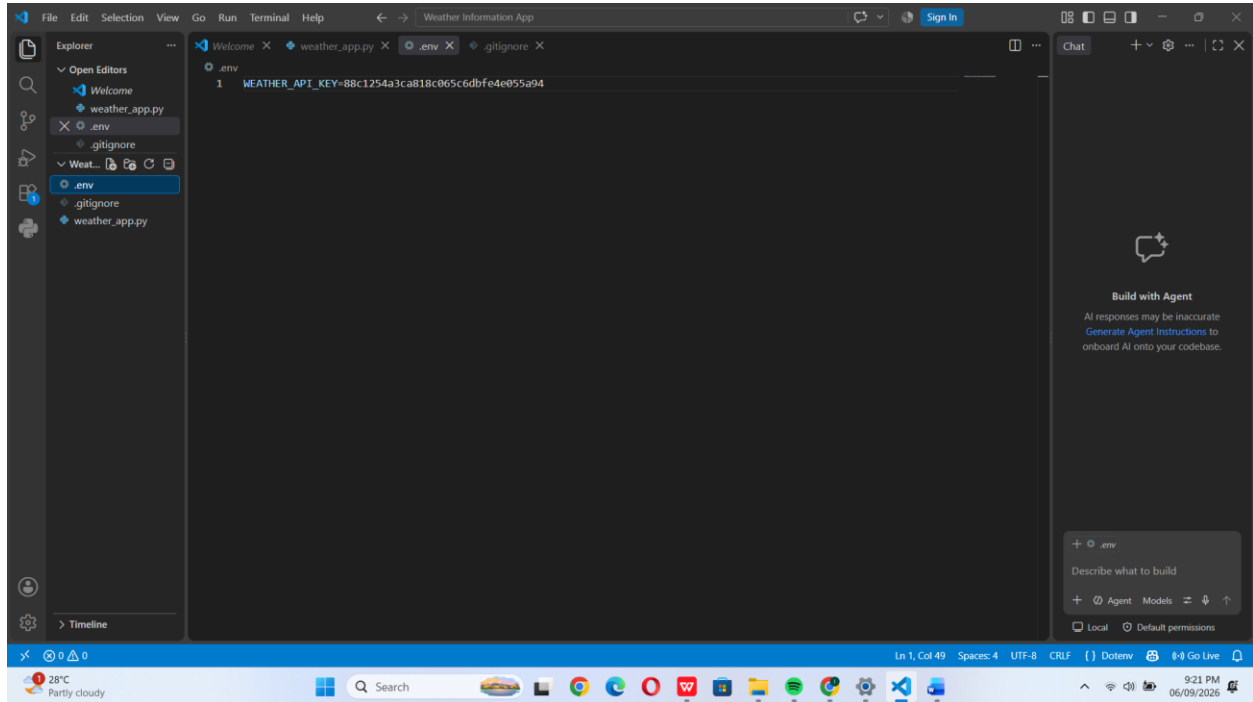
Step 3: Save the program as:

```
weather_app.py
```

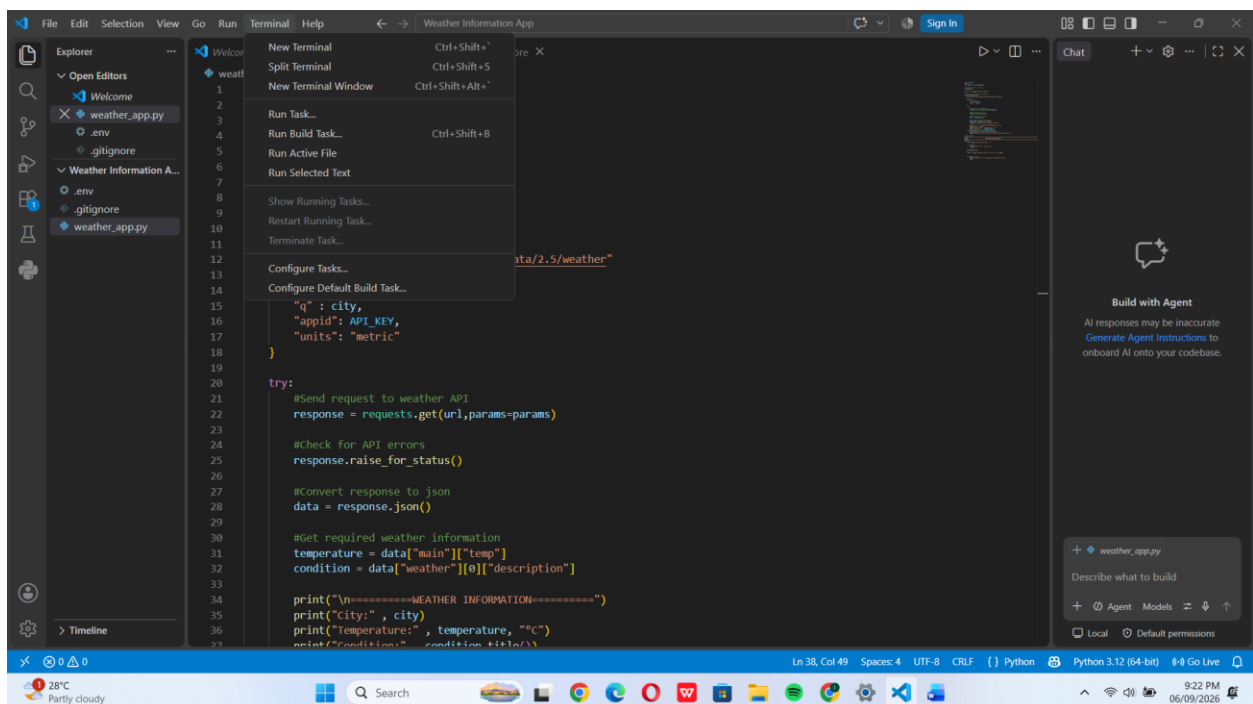
Step 4: Create a .env file in the same project folder.

Add your API key:

Weather_API_KEY = your_api_key_here

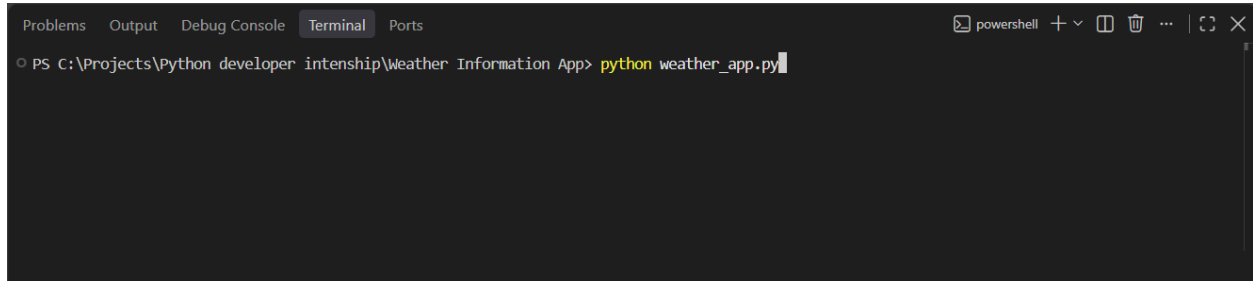


Step 5: Open Command Prompt / Terminal -> New Terminal



Step 6: Run the program:

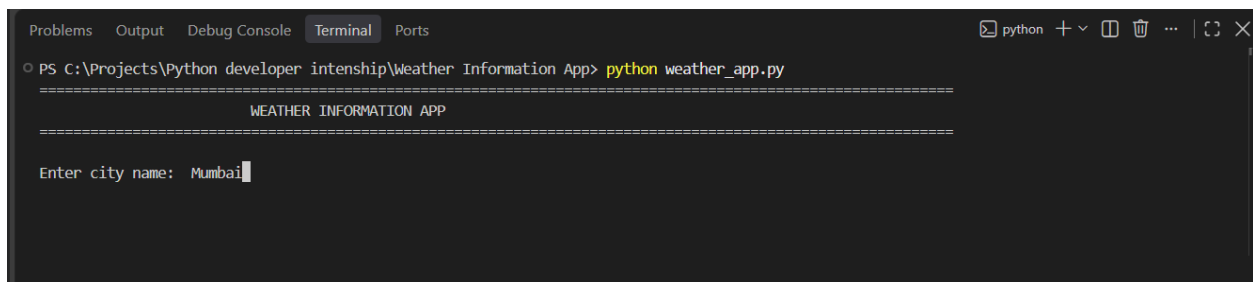
```
python weather_app.py
```



```
Problems Output Debug Console Terminal Ports
PS C:\Projects\Python developer intenship\Weather Information App> python weather_app.py
```

Step 7: Enter a city name.

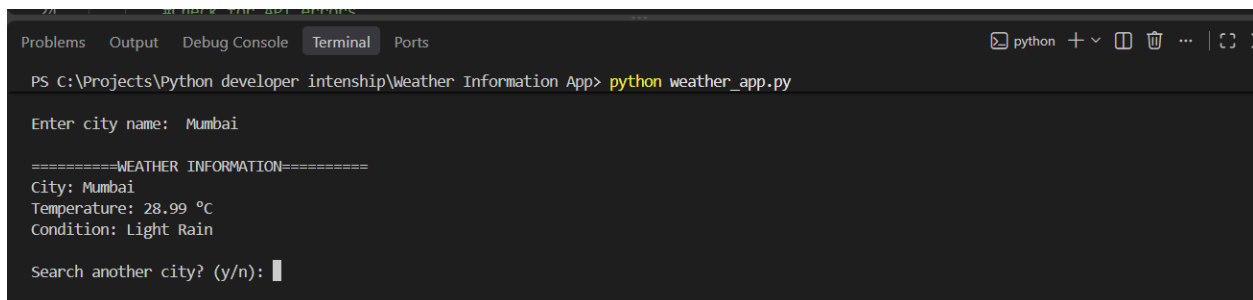
Enter city name: Mumbai



```
Problems Output Debug Console Terminal Ports
PS C:\Projects\Python developer intenship\Weather Information App> python weather_app.py
=====
WEATHER INFORMATION APP
=====

Enter city name: Mumbai
```

Step 8: The application displays the current weather information.



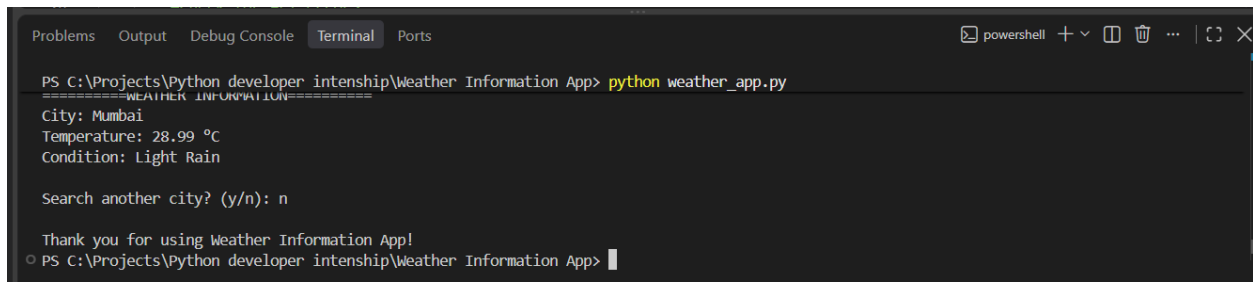
```
Problems Output Debug Console Terminal Ports
PS C:\Projects\Python developer intenship\Weather Information App> python weather_app.py

Enter city name: Mumbai

=====WEATHER INFORMATION=====
City: Mumbai
Temperature: 28.99 °C
Condition: Light Rain

Search another city? (y/n):
```

Step 9: Enter **y** to search for another city or **n** to exit.



The image shows a terminal window with a dark background. At the top, there are tabs for 'Problems', 'Output', 'Debug Console', 'Terminal', and 'Ports'. The 'Terminal' tab is active. The prompt is 'PS C:\Projects\Python developer intenship\Weather Information App>'. The user has entered 'python weather_app.py'. The output of the script is as follows: a separator line of equals signs, the text 'City: Mumbai', 'Temperature: 28.99 °C', 'Condition: Light Rain', a prompt 'Search another city? (y/n): n' where 'n' has been entered, a 'Thank you' message, and a final prompt 'PS C:\Projects\Python developer intenship\Weather Information App>'.

```
PS C:\Projects\Python developer intenship\Weather Information App> python weather_app.py
=====WEATHER INFORMATION=====
City: Mumbai
Temperature: 28.99 °C
Condition: Light Rain

Search another city? (y/n): n

Thank you for using Weather Information App!
PS C:\Projects\Python developer intenship\Weather Information App>
```

Result: The Weather Information App successfully retrieves and displays real-time weather information using a REST API.